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Single-Wire Time Domain Reflectometry Technique (SW-TDR): Detecting Faults in Power System Grounding Electrodes

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ABSTRACT The performance of electrical power systems relies on a healthy and properly functioning grounding network. Buried vertical electrodes are the pillars of a grounding system. This paper presents a Time Domain Reflectometry (TDR) technique based on surface wave propagation along a single wire to detect a fault. In SW-TDR, a fast rise-time pulse is injected onto the single conductor grounding electrode primarily exciting transverse magnetic (TM) mode surface wave propagation. The surface wave propagates along the electrode and is reflected at any impedance mismatch such as a fault in the electrode. The mismatch location and severity of the fault can be identified using the reflected signal waveform. Expressions for the fields of the surface wave supported by a single electrode in a lossy media is presented. Full wave electromagnetic simulation is used to evaluate the wide-band input impedance and then FFT is applied to determine the TDR response. Simulation results show that SW-TDR can identify a break-point or even partial corrosion of a grounding electrode for a wide range of soil conductivity for a system bandwidth of 200 MHz. A surface wave launcher design is also presented which enables the SW-TDR to be implemented without disconnecting the electrode from the grounding grid. A scale model experiment demonstrates the feasibility of the SW-TDR approach. Measurements show detection capabilities are similar to those obtained by simulation.

INDEX TERMS Grounding electrode corrosion, fault detection, fault location, surface wave propagation, single conductor transmission line, time domain reflectometry (TDR)

I. INTRODUCTION

The health status of a grounding network needs to be monitored frequently for safe and efficient operation of power system substations as well as for large industrial operations [1]–[3]. A grounding network consists of vertical and horizontal conductors. The purpose of the horizontal conductors is to interconnect the vertical electrodes and the vertical electrodes provide a path for current produced during a lightning strike or fault in the electrical equipment. The vertical ground electrodes will only work efficiently if they form a low impedance path to the ground. A break-point is a worst-case scenario which essentially shortens the electrode's length. Vertical electrodes can show significant corrosion after 10-20 years of installation and can be corroded up to 80% in that period of time [4], [5]. This corrosion is the prime reason that renders the grounding system ineffective and can lead to critical electrical equipment damage or personal safety risk [6]–[8] and is responsible for an enormous economic loss [5], [9]. At present health monitoring of grounding networks are

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infrequent, approximately once in 20-50 years [5], [10]. Currently physical examination which involves electrode excavation is the only method of diagnostics, and is time consuming, expensive, and inefficient. Most power substation grounding systems have not been inspected because of the absence of technology that can do inspection in real-time with low cost. This results in a potential threat not only for worker safety but also for non-disrupted power transmission. Therefore, development of new cost-effective technologies for detecting faults in a grounding system is of great interest.

Most of the research on grounding system fault detection has been developed using circuit theory techniques. The simplest approach involves measuring the DC or low frequency resistance between two points in a grounding network and relating the health of the tested area to the increment or decrement of the measured resistance. This method has been advanced by applying different circuit algorithms, matrix theory and network theory [7], [9]–[13] which help to provide an overall health status of a grounding system or part of the system that is in under test. The circuit methods are not a good candidate for grounding system fault detection in terms of locating a specific electrode fault position or its severity. The resistance measurement is highly influenced by the soil resistivity, hence a small increment due to a single fault is hard to identify. Furthermore, precise topology of the grounding system needs to be known in order to apply different circuit network theories and algorithms.

Other approaches have been proposed, such as electromagnetic field methods. In the electromagnetic field method, the magnetic field distribution on the surface of the grounding grid is measured upon injecting a DC or low frequency current. The measured magnetic field will appear distorted in the defected region which is an indication of a break-point. To analyze the distorted magnetic field due to corrosion damage various imaging and tomography techniques have been utilized [14]–[19]. In spite of having advantages over circuit-based methods, the electromagnetic field method has drawbacks. The magnetic field measurement is highly susceptible to outside electromagnetic interference leading to inaccurate results [11], [12]. Also, for conducting the measurement and interpreting the measured results, precise topology of the grounding grid has to be known beforehand. Recently, a TDR approach using time reversal theory was proposed for detecting faults on horizontal grounding electrodes [20]. Here the horizontal grounding conductor embedded below the earth-air interface is treated as a lossy transmission line. Alternatively, an acoustic guided wave approach has drawn attention for grounding electrode fault detection. The method is based on injecting an acoustic wave onto the electrode which reflects from a fault and carries information on the fault location with it [21], [22]. However, launching the acoustic wave is challenging as it relies on proper contact between the transducer and ground electrode and reflections from the grounding grid must be considered.

The current electromagnetic-based methods for ground electrode health diagnostics are unable to precisely identify fault location or provide an estimate of fault severity. In this paper a new single-wire time domain reflectometry (SW-TDR) technique, based on surface wave propagation, is introduced. It is shown how the SW-TDR can be applied to grounding electrode fault detection. The new technique can identify both complete break-points or severely corroded regions of the electrode. Section II of the paper introduces the SW-TDR technique and section III develops a theory based on surface wave propagation. In section IV and V, full-wave electromagnetic simulation is used to study different grounding electrode scenarios. In section VI a surface wave launcher design is introduced. Finally, in section VII results of a scale-model experiment are presented, demonstrating the SW-TDR technique.

II. SINGLE-WIRE TIME DOMAIN REFLECTOMETRY

Time domain reflectometry (TDR) is a widely used method in electrical system diagnosis, material measurement, soil science and structural health monitoring [23]–[26]. Because of its simplicity and versatility it has the potential to be used in many applications [24], [27], [28].

Conventional TDR uses two conductors so that a transverse electromagnetic (TEM) wave is supported. For single grounding electrode fault detection an additional electrode would be required to perform TDR measurement. Such a case is investigated in [29], [30], where a secondary conductor is inserted next the electrode to support a TEM-wave. Alternatively, for a single conductor grounding electrode, surface wave excitation can be employed for TDR fault detection. In the SW-TDR technique a transverse

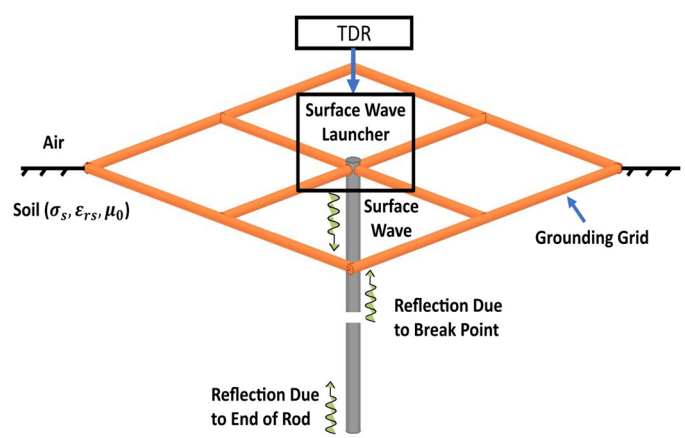

FIGURE 1. Schematic representation showing the working principle of the SW-TDR method. The grounding system consists of a single vertical electrode tied to a grounding network. This is embedded in a non-magnetic, homogeneous soil with conductivity, σ_s , and relative permittivity, ε_{rs}.

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magnetic (TM) mode surface wave is injected onto the grounding electrode. Similar to conventional TDR, a fast rise-time pulse is injected onto and propagates along the electrode and is reflected from any impedance discontinuity, such as a break-point, as shown in Fig. 1. The location of the impedance mismatch can be extracted from the captured reflected signal travel time given that the soil electrical parameters are known [31].

Time domain reflectometry using surface wave sensors has previously been used for power line fault detection [32], structural health monitoring [33] and soil water content measurement [34], [35].

III. SURFACE WAVE PROPAGATION ON A CONDUCTOR IN LOSSY MEDIA

Surface wave propagation on a single lossy conductor in free space was investigated by Sommerfeld [36], who formulated a solution for the transverse magnetic symmetric mode of propagation. Later, Goubau studied how this single wire mode could be used for low-loss communication and various launcher strategies were developed [37]–[40]. Olsen presented a detail formulation of an infinite wire in air excited by a gap source or elementary dipole source next to the wire [41]. A spectral domain approach was used and showed that a complete solution for an infinite wire, but finite source excitation, should include both radiation and discrete surface wave mode contributions. The input impedance of a finite length wire embedded vertically in a lossy half-space medium has been extensively studied, usually in the context of a grounding electrode and for lengths much smaller than the wavelength in the earth medium [42], [43]. When a short wire source is buried in a lossy half-space earth medium, contributions from the TM surface wave along the wire, the radiated fields, as well as a radial propagating surface wave along the air-earth interface should be considered [44]. Full wave integral-equation and three-dimensional numerical techniques have been used for determining the impedance, current distribution and transient response of a single vertical wire embedded in a lossy earth [45]–[49].

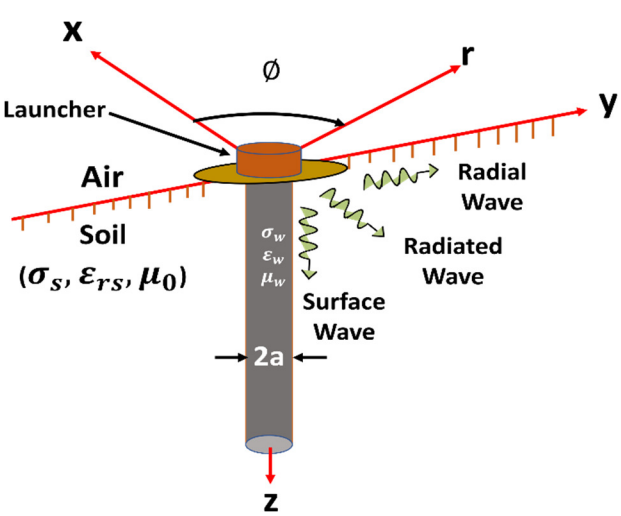

FIGURE 2. Single conductor buried in a non-magnetic, homogeneous soil with conductivity, σ_s , and relative permittivity, ε_{rs}. A launcher at $z = 0$ will excite several propagating modes. A conductive screen (ground mat) can be placed around the launcher to act as a ground plane.

In this paper, numerical simulation is employed to study transient pulse wave propagation along a finite length single-wire buried in lossy half-space. Studies concentrate on situations where the pulse frequency bandwidth is high enough such that the wire is several wavelengths long. This single-wire TDR configuration is applied to detecting partial faults or complete break-points in a buried grounding electrode. A vertical grounding electrode buried in soil can be considered as a cylindrical conductor embedded in a lossy dielectric as shown in Fig. 2. The launcher and finite length conductor will support a radiating wave component, a radial propagating surface wave bound to the air-earth interface as well as a surface wave propagating along the conductor. All these wave components are incorporated into the full wave numerical solutions, but the excitation of each one depends on the launcher design. Here, the axial wave bound to the conductor (Sommerfeld TM-surface wave) is derived since the fundamental azimuthal symmetric mode has small attenuation and thus it has potential to detect any impedance mismatch along the electrode. To determine the propagation characteristics of the surface wave the conductor is initially assumed to be infinitely long (the air-earth interface is neglected or alternatively there is a ground plane at $z = 0$). A thin-wire approximation is used in the analysis. Fig. 2 shows the grounding electrode geometry where a is the radius. The earth region, $z > 0$ and $r > a$, is characterized by conductivity, σ_s, relative permittivity, ε_{rs}, and permeability, μ_0. The conductor, $z > 0$ and $r \leq a$, is characterized by conductivity, σ_w, permittivity, ε_{rw}, and permeability, μ_0. The fields generated inside and outside the conductor are found by solving the wave equation in each region and then satisfying the boundary conditions at their interface [50]. Assuming an $e^{+j\omega t-\gamma z}$ time and axial dependence, the resulting electric and magnetic fields in the conductor $r \leq a$, can be expressed as:

$$\bar{E}^w(r) = \frac{I}{2\pi\sigma_w a}\left[\gamma\frac{J_1(T_w r)}{J_1(T_w a)}\hat{r} + T_w\frac{J_o(T_w r)}{J_1(T_w a)}\hat{z}\right]e^{-\gamma z}, \quad (1)$$

$$\bar{H}^w(r) = \frac{I}{2\pi a}\left[\frac{J_1(T_w r)}{J_1(T_w a)}\,\hat{\phi}\right]e^{-\gamma z}. \quad (2)$$

Similarly, in the surrounding earth, $r > a$, the fields are:

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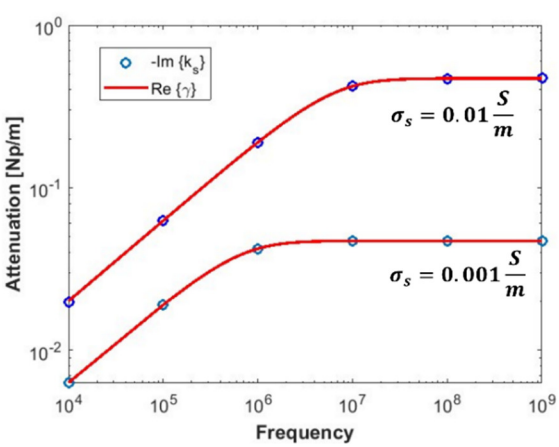
$$\bar{E}^s(r) = -\frac{j\omega\mu_o I}{2\pi k_s^2}\left[T_s\gamma\frac{H_1^{(2)}(T_s r)}{(T_s a)H_1^{(2)}(T_s a)}\hat{r} + T_s^2\frac{H_0^{(2)}(T_s r)}{(T_s a)H_1^{(2)}(T_s a)}\hat{z}\right]e^{-\gamma z}, \quad (3)$$

$$\bar{H}^s(r) = \frac{I}{2\pi}\left[T_s\frac{H_1^{(2)}(T_s r)}{(T_s a)H_1^{(2)}(T_s a)}\hat{\phi}\right]e^{-\gamma z}. \quad (4)$$

In the above, $k_s = \sqrt{\omega^2\mu_0\varepsilon_0\varepsilon_{rs} - j\omega\mu_0\sigma_s}$ and $k_w = \sqrt{-j\omega\mu_0\sigma_w}$ are the wavenumbers in the earth and wire medium, respectively. I is a source current, γ is the complex propagation constant, J_o and J_1 are Bessel functions of first kind, and $H_0^{(2)}$ and $H_1^{(2)}$ are Hankel functions of second kind. Here, $T_s = \sqrt{\gamma^2 + k_s^2}$ and $T_w = \sqrt{\gamma^2 + k_w^2}$, with $Im\{T_S \leq 0\}$ and $Im\{T_w \leq 0\}$, are the transverse wavenumbers in the earth and wire medium, respectively. The goal is to determine the complex propagation constant γ for the fundamental TM mode. Imposing boundary condition, $E_z^w(a) = E_z^s(a)$, at the surface of the conductor, $r = a$, results in the eigenvalue equation:

$$\frac{T_w}{2\pi\sigma_w a}\frac{J_o(T_w a)}{J_1(T_w a)} = -\frac{j\omega\mu_g}{2\pi k_s^2}T_s^2\frac{H_0^{(2)}(T_s a)}{(T_s a)H_1^{(2)}(T_s a)} \quad . \quad (5)$$

The complex propagation constant, γ, is obtained by solving this equation. The attenuation, Re $\{\gamma\}$, and phase, Im $\{\gamma\}$, constants are shown in Fig. 3a and 3b, respectively, for typical earth parameters along with the earth wavenumber, k_S. The fundamental TM mode propagation constant is nearly the same as the earth wavenumber $\gamma \approx jk_s$. The complex radial wavenumber, T_S, is subsequently small as shown in Fig. 3c. Using small argument approximation, $|T_s| \ll 1$, for the Hankel functions in eqn. 3 and 4, the field are expected to have a



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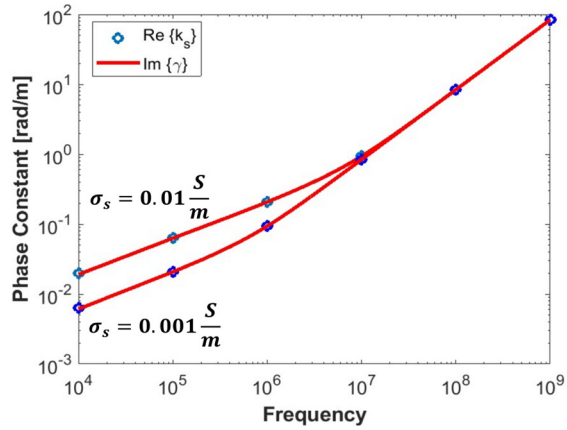

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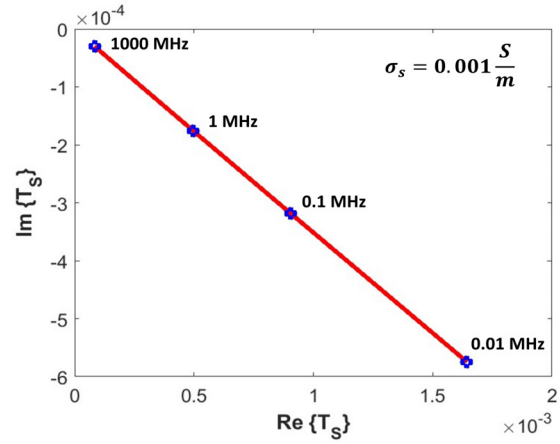

(c)

FIGURE 3. (a) Attenuation and (b) phase constant of the fundamental TM mode evaluated by solving (5). The propagation constant is similar to the wavenumber in earth medium (c) Transverse wavenumber, T_S, showing that the radial wavelength, $\lambda_r = 2\pi/Re\{T_S\}$, is large. The soil permittivity is $\varepsilon_{rs} = 16$ is these cases.

$1/r$ behavior near the conductor. Far from the conductor, $|T_s| \gg 1$, fields decay exponentially and have a radial wavelength, $\lambda_r = 2\pi/Re\{T_S\}$, which is greater than 100m for the cases that will be studied.

Note that, the soil exhibits dispersionless characteristics after 30 MHz (constant attenuation and linear phase). As such the SW-TDR pulse bandwidth is chosen so that most of the pulse energy falls into 30-200 MHz as in Fig. 3a. This helps to maintain the shape of the pulse, albeit with higher loss, helping to identify the fault location and severity by using reflected pulse timing and shape.

IV. FULL WAVE SIMULATION

The TDR instrument injects a fast rise-time pulse onto the single wire transmission line and measures reflections due to impedance discontinuities in the line. In this paper a full wave simulator, High Frequency Structure Simulation

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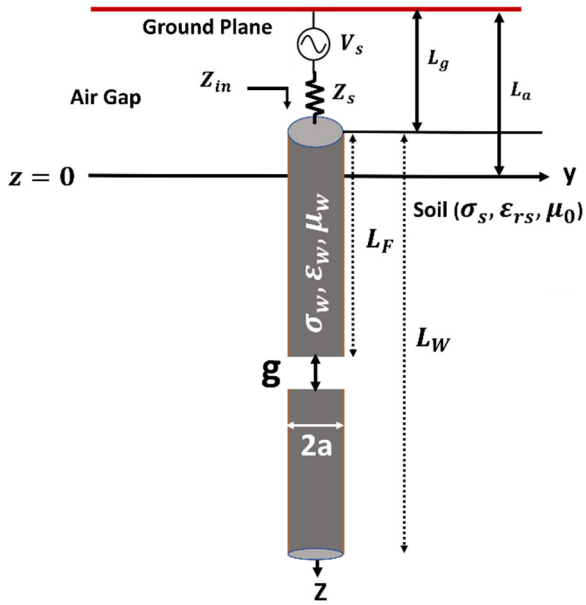

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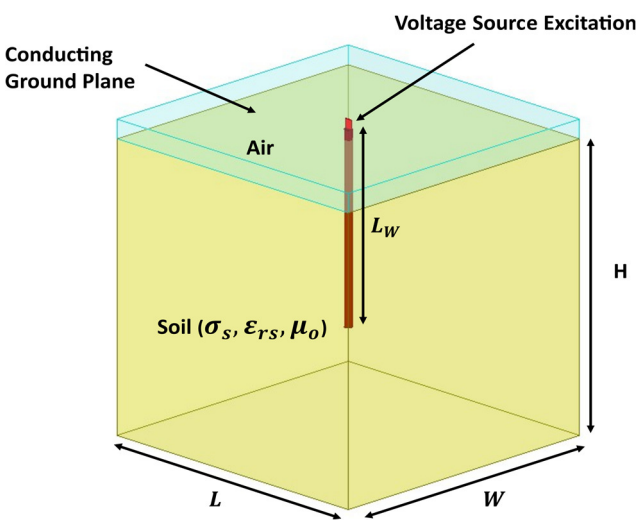

(b)

FIGURE 4. (a) Full wave simulation model of the single-wire TDR. An air gap with a ground plane is included above the soil to connect the launcher to the buried electrode. (b) Three- dimensional geometry used for finite element numerical simulation of the grounding electrode.

(HFSS) [51], is used to determine the wideband input impedance, Z_{in}, of the grounding electrode in the frequency domain. To excite the electrode, a gap excitation between a large ground plane (mimicking the grounding network or grounding mat placed on the surface) is used. A harmonic voltage source is applied to the electrode for a frequency range of 0 to 300 MHz. Fig. 4a shows the simulation model of the SW-TDR detection system. Here, L_W is the length of the grounding electrode, L_F is the distance to the fault location, g is the break-point gap size and a is the radius of the electrode. L_a is the air region and L_g is the gap between the electrode and ground plane for excitation. The excitation is implemented using a lumped port in HFSS. Once the input impedance is determined the voltage at input terminal is obtained as:

$$V_{in}(\omega) = \frac{Z_{in}(\omega)}{Z_{in}(\omega) + Z_s} V_s(\omega) \ . \tag{6}$$

Equation (6) is a sum of incident and reflected surface wave fields but also includes the ground radial wave and radiated wave components. $V_s(\omega)$ is the Fourier transform of the input pulse and Z_s is the TDR source impedance. TDR instruments typically measure a step response. A 200 MHz band-limited step response is used in this paper. The inverse Fourier transform of (6) provides the impulse response in time domain. In this paper, most results are presented in the form of impulse response as it provides direct estimation of return pulse travel time and can be obtained by differentiating the step response. For lossy dispersionless propagation, the location of a fault can be identified from the response waveform as:

$$L_F = \frac{v\Delta T}{2}, \tag{7}$$

where ΔT is two-way time delay of an injected pulse and v is the wave velocity in the soil medium.

A. WAVE PROPAGATION STUDY

In this section the fields supported by a single buried electrode are determined using HFSS simulation and are compared with TM-wave surface wave theory. The configuration presented in Fig. 4 with no break-point ($g = 0$) is examined for soil conductivity $\sigma_s = 0.001\ S/m$, permittivity $\varepsilon_{rs} = 16$ and permeability $\mu_0 = 1$ and for electrode dimensions, $L_W = 2$ m, $L_a = 20$ mm, $L_g = 5$ mm and $a = 10$ mm. Fig. 5a shows the simulated time-average Poynting vector around the conductor for 100 MHz. Near the conductor the Poynting vector is primarily axially directed. Far from the conductor and near end of the line significant radial components are evident, indicating a significant radiation mode is excited. Additionally, a radial wave at earth-airgap-ground plane interface is observed. The scaled axial component of the time-average Poynting vector along the conductor and for different radial distances is shown in Fig. 5b (scaled by a r^2 factor). The axial component is the same for different radial distance, indicating that the fields decay as $1/r^2$. This is congruent with the radial electric field and azimuthal magnetic field given by the TM-mode theory by equations (3) and (4) in the small argument limit. The decay of the axial Poynting vector is not exponential, indicating both forward and reflected waves as well as a radiating component exists. Fig. 5c shows the radial component of the Poynting vector (scaled by a r^2 factor). The radial component is significantly less than the axial component except at the excitation point where a radial wave at the ground plane-soil interface is supported, and at the end termination where reactive near-field and radiated components are generated. Results of Figs. 5b and 5c are

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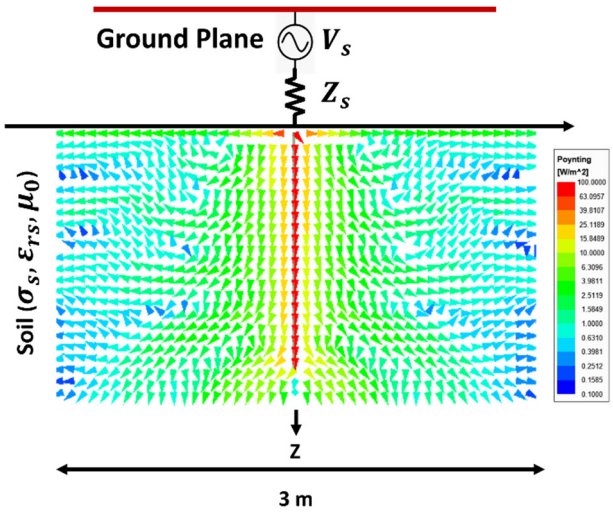

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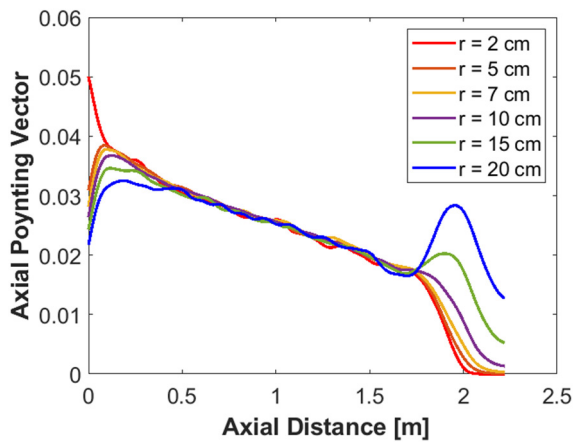

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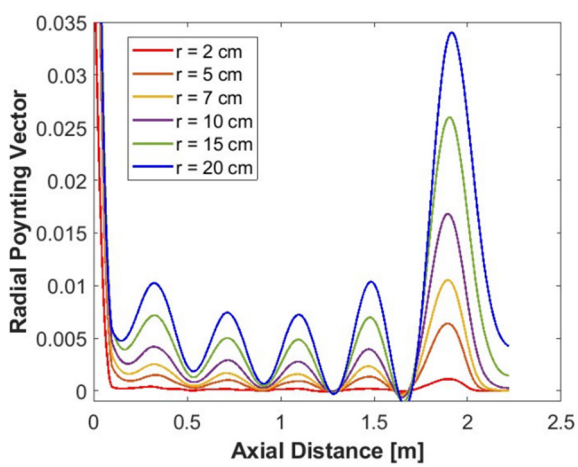
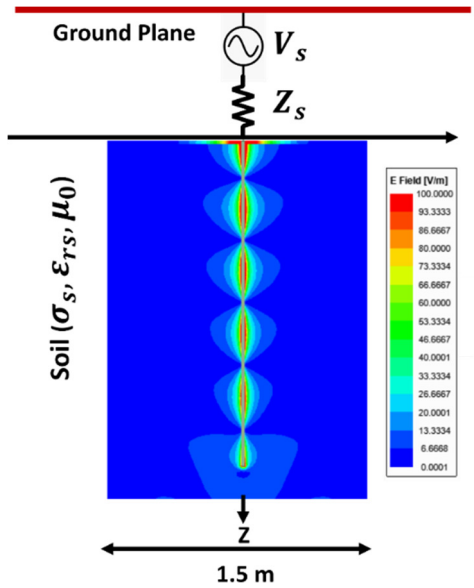


(c)

FIGURE 5. (a) Time average Poynting vector at 100 MHz (b) Axial component of Poynting vector along the conductor for different radial distances scaled by r^2 (c) Radial component of Poynting vector for different radial distances scaled by r^2.



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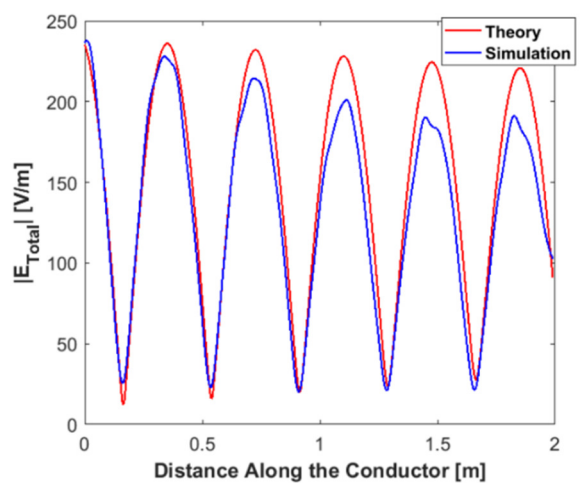

(b)

FIGURE 6. (a) Magnitude of the electric field for a snapshot in time domain at 100 MHz showing a standing wave behavior (b) Comparison between TM mode theory (eqn. 8) and simulation of total electric field near the surface of the conductor.

similar to those observed for a long monopole antenna [52] in air (except the conductor is embedded in soil).

Figure 6a shows the simulated electric field magnitude for a source frequency of 100 MHz. The standing wave pattern observed is due to reflection at the end of the electrode. Fig. 6b shows the total electric field near the surface of the conductor, r=2 cm, for both the simulated and TM-mode theory-based approximation. Assuming a TM surface wave, the standing wave field can be approximated as,

$$|E(z)| = \left|E^i e^{-\gamma L_w}(e^{-\gamma(z-L_w)} + \Gamma_L e^{+\gamma(z-L_w)}\right|, \qquad (8)$$

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where E^i is the incident field, Γ_L is a reflection coefficient at the end of the electrode and γ is the propagation constant obtained from (5). The full wave simulation result follows the behaviour of the TM mode assumption, equation (8), also showing a strong standing wave behavior. Differences are due to the radiation mode contribution, which also has a strong axial Poynting vector component (similar to the behavior of a long wire in air [52]).

B. FAULT DETECTION USING SW-TDR

In this section, fault detection using the SW-TDR response is demonstrated. This paper targets identifying complete break-points or severely corroded regions on a grounding electrode that might cause a reduced radius in the corrosion region. The propagation constant of the pulse is dominated by the surrounding earth region dielectric parameters. Thus, the possible formation of a thin corrosion layer on the electrode surface is not considered as the TDR signal shape and attenuation is not severely affected by its presence. Primarily, reflections will only be seen due to severe faults such as break-points or severely reduced radius regions. This model for a ground electrode has been used by other researchers [20], [53]–[55].

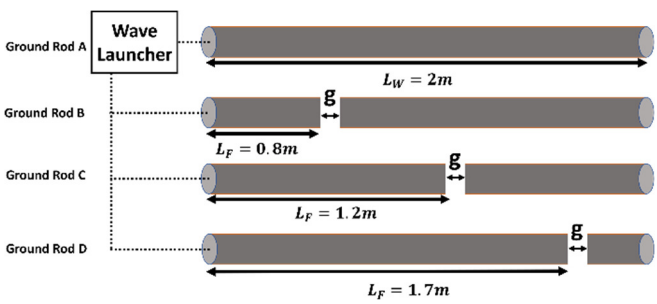

FIGURE 7. Schematic of ground rod defect scenarios with a break-point, g, at different locations.

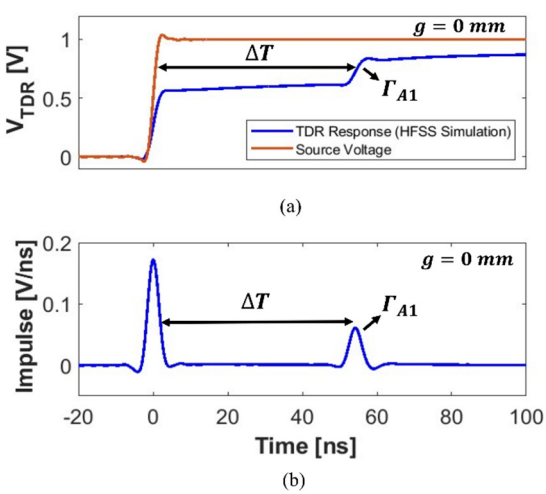

FIGURE 8. (a) Simulated TDR step and (b) impulse responses for a defect-free ground rod, case A (a = 10 mm, L_W = 2 m, g = 0 mm, $\sigma_s = 0.001$ S/m, $\varepsilon_{rs} = 16$).

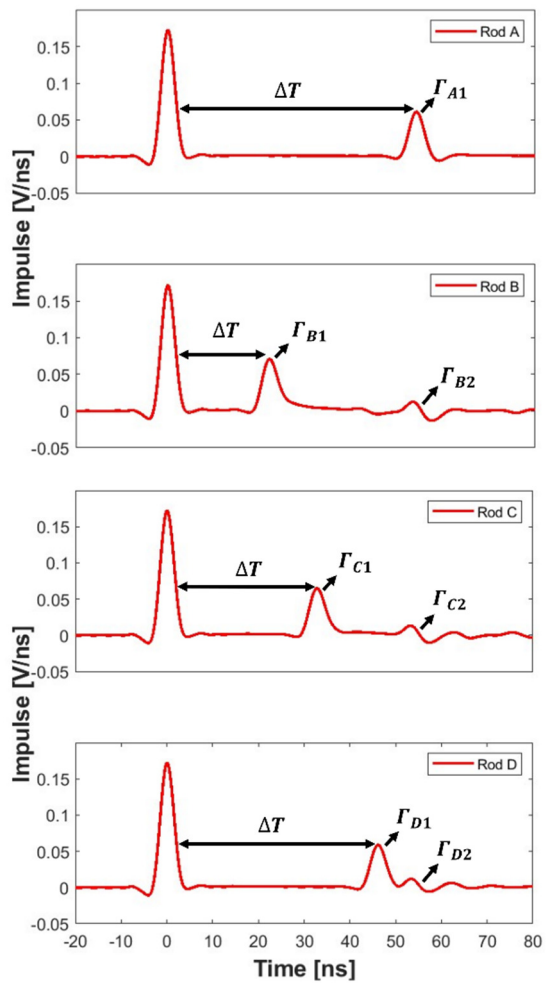

FIGURE 9. Impulse responses for ground rod fault cases A, B, C and D as described in Fig. 7 (a = 10 mm, L_W = 2 m, g = 20 mm, $\sigma_s = 0.001$ S/m, $\varepsilon_{rs} = 16$).

In our numerical simulation the grounding electrode is placed in a 2 m × 2 m × 3 m simulation space as shown in the Fig. 4b. There are two regions in the model: soil and air. All boundaries are truncated with a numerical radiation boundary condition, except for the top side of the simulation region which is a perfect electric conducting (PEC) boundary. A voltage source is placed between the PEC and the grounding electrode and Maxwell's equations are solved numerically using the finite element method in HFSS [56]–[58] to obtain the wideband input impedance $Z_{in}(\omega)$ in the frequency domain. The impulse response is obtained from equation (6) using the FFT and then numerical integration is applied to provide the step response. Past studies of grounding systems have considered electrodes of length 2-10 m [48], [56], [59], [60]. In this study, a 2-meter rod is used with a diameter of 20 mm. The electrode diameter has minimal influence on the grounding electrode impedance [48] and will not affect the TDR pulse propagation characteristics appreciably as it is dominated by the soil

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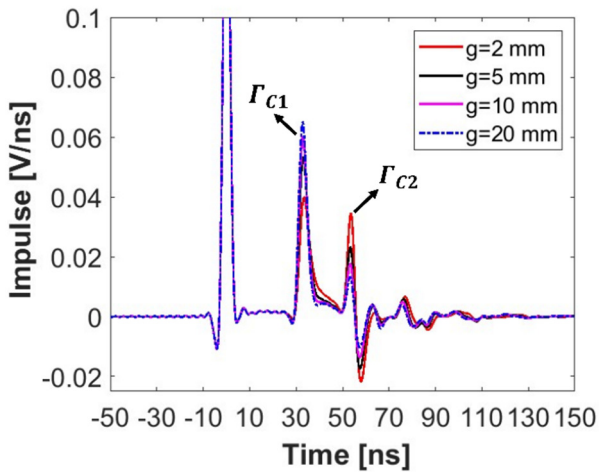

FIGURE 10. Impulse response for different break sizes, g, for case C (see Fig. 7) with a = 10 mm, L_W = 2 m, $\sigma_s = 0.001$ S/m, and ε_{rs} = 16.

TABLE I
CALCULATED AND MEASURED TIME DELAYS OF THE REFLECTED PULSE

	Break-point, ΔT [ns]		End of Rod, ΔT [ns]	
	Theoretical	Simulated	Theoretical	Simulated
Rod A	n/a	n/a	53.40	54.00
Rod B	21.30	22.20	53.40	53.40
Rod C	32.05	32.80	53.40	53.40
Rod D	45.40	46.18	53.40	53.60

electrical parameters. A shorter length electrode is used as this is a worse-case situation when trying to discriminate fault locations from the TDR response signal.

Four ground rod fault scenarios A, B, C and D are considered as shown in Fig. 7. Rods that are defective have a break-point, g = 20 mm. Rod A has no break-point, with $g = 0$, Rods B, C and D have faults located at 0.8 m, 1.2 m and 1.7 m, respectively. Since Rod A is defect free, only a reflection from the end of the line is expected. Fig. 8a and 8b show simulated TDR step and impulse responses respectively for test Rod A. The input impulse occurs at 0 ns and the reflection due the end of the rod, marked with Γ_{A1}, is at 54 ns. The simulated impulse responses for case B, C and D are shown in Fig. 9. The reflection produced by the break-points are identified by Γ_{B1} at 22.20 ns, Γ_{C1} at 32.80 ns and Γ_{D1} at 46.18 ns, respectively. Reflection from the end of the electrode is identified by Γ_{B2}, Γ_{C2} and Γ_{D2} at 53.40 ns. The two-way time delay, ΔT, can be used to determine the location of the faults using (7).

Table I lists the time delays for the reflected pulses generated from break-points and the end of the rod for the cases presented in Fig. 7. The theoretical delay is calculated from equation (7), where the wave velocity is estimated based on the soil electrical parameters ($\sigma_s = 0.001$ S/m, and $\varepsilon_{rs} = 16$). The reflected pulse time delays are obtained using the simulation results in Fig. 9.

Not only can the location of a fault be determined, but the severity of a fault in the grounding electrode can be deduced by analyzing the SW-TDR response waveform. Fig. 10 shows the TDR signals for different break-point size, g, for case C in Fig. 7. A stronger reflection will be generated by the forward traveling wave for a larger gap, g, at the fault, Γ_1. Comparison of the magnitudes of reflected waves due to fault, Γ_1, and from the end of the line, Γ_2, can provide fault severity details.

C. INFLUENCE OF SOIL CONDUCTIVITY

Soil permittivity and conductivity characteristics depend on environmental conditions, water content and may vary with depth and season related temperature. The electrical parameters for different types of soil can have a wide range of values [43], [48], [56], [59], [61]. For the TDR bandwidth of the system, soil permittivity influences the propagation delay but does not influence the TDR pulse attenuation (see Fig. 3). Soil conductivity, however, plays a role in TDR response signal attenuation. Therefore, one permittivity value and several conductivity values are used in the case studies. The attenuation of the TDR pulse as it travels along the electrode depends strongly on the conductivity of the soil medium and is described by the attenuation constant in Fig. 3a for different soil conductivities. Strong attenuation can dramatically affect the SW-TDR system's ability to detect a fault. Fig. 11 shows the impulse response for case C (see Fig. 7) for different soil conductivity. Soil conductivity values are chosen based on those commonly used in grounding system studies [43], [48], [56], [59], [61]. For a TDR pulse with bandwidth 200 MHz, when $\sigma_s \leq 0.001$ S/m, the wave attenuation is small and a large reflection due to the fault is observed. For soil conductivity $\sigma_s > 0.01$ S/m, the forward and reflected waves decay faster as they propagate. These studies show the SW-TDR method works for wide range of soil conductivity, but it is more efficient in low loss or moderate lossy soil conditions. Note that even though the

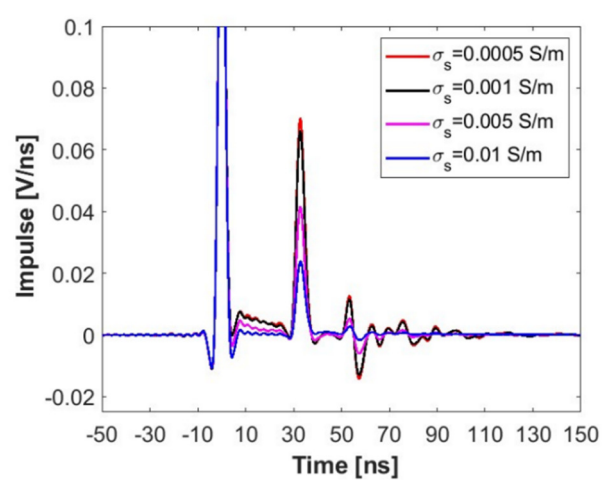

FIGURE 11. Impulse response of the SW-TDR system showing the reflected pulse amplitude for different soil conductivity with a = 10 mm, L_W = 2 m, L_F = 1.2 m, g = 20 mm, and ε_{rs} = 16.

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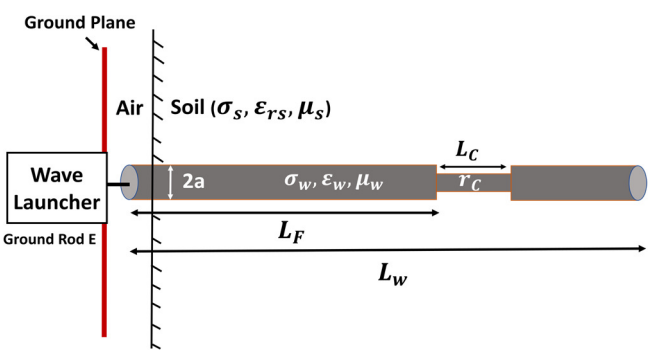
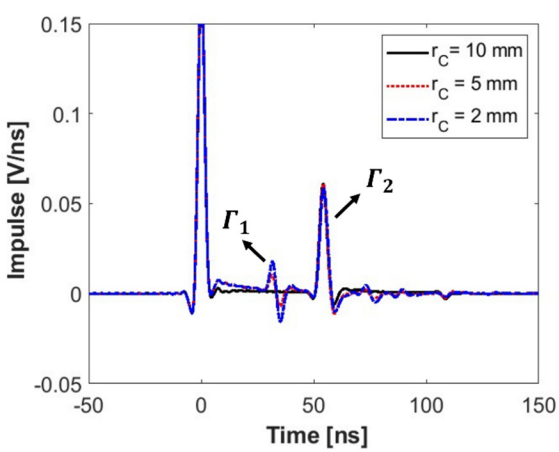
reflected signal amplitude is attenuated for higher conductivity soil conditions, the narrow pulse shape is maintained. Thus, a high-amplitude TDR pulse can be used to enhance the reflected pulse signal-to-noise ratio in the higher loss situations, without sacrificing fault detection capability.

V. DETECTION OF PARTIAL CORROSION

In previous sections, faults caused by a complete break-point were analyzed, which is the most severe situation affecting grounding electrode performance. Before a complete break-point occurs a reduction in radius of the electrode due to a localize corrosion occurs over a long period of time. The possibility of detecting such reduction is investigated by examining grounding electrode E in Fig. 12a. Here the reduced radius, r_C , section with length, L_c, models the partially corroded region. Fig. 12b gives the impulse response for different values of r_C showing that the detection of a partial fault is possible. The reflected pulse, Γ_1, generated from the reduced part of the electrode is bipolar since the reflection results due to a lower to higher impedance discontinuity at the start of the fault section and then from a higher to lower impedance discontinuity at its end. This result is similar to that obtained for a two-wire TEM TDR detection system where partial fault is examined [29].



(a)

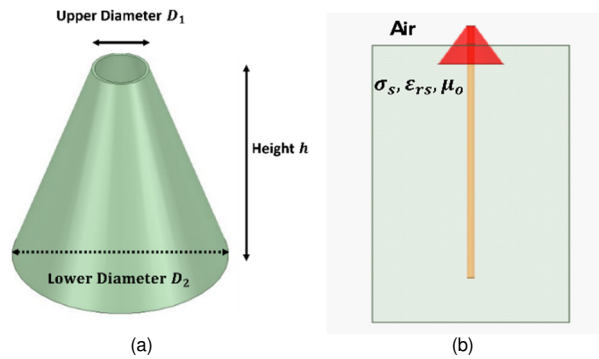


(b)

FIGURE 12. (a) Schematic of a ground rod with localized partial radius reduction due to corrosion, and (b) Impulse responses for this case showing the reflected signals for three different partial radius, r_C, values. The length of the reduce radius section is $L = 100$ mm. Γ_1 is the reflection due to the corroded section and Γ_2 is the reflection due to end of the line. (a = 10 mm, $L_w = 2$ m, $L_F = 1.2$ m, $\sigma_s = 0.001\ S/m$, $\varepsilon_{rs} = 16$).

VI. FAULT DETECTION WITH A LOCALIZED LAUNCHER

In the previous sections a large ground plane or ground mat with gap excitation is used. This would entail cutting the



(a) (b)

FIGURE 13. (a) Conical horn launcher concept and (b) excitation of a grounding electrode using the launcher.

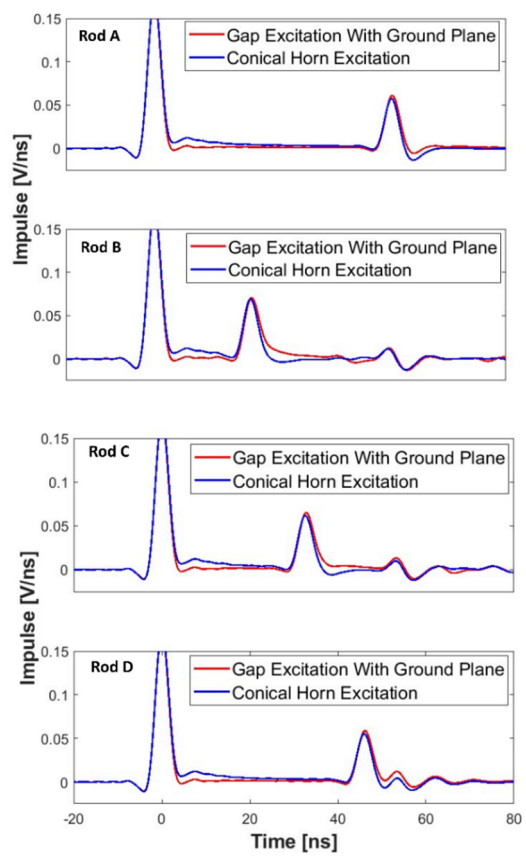


FIGURE 14. Impulse responses of ground rod cases A, B, C, and D (see Fig. 7) using ground-plane-gap excitation and conical horn launcher SW-TDR design. ($a = 10$ mm, $L_W = 2$ m, $g = 20$ mm, $\sigma_s = 0.001$ S/m, $\varepsilon_{rs} = 16$).

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grounding electrode at the surface of the soil to disconnect it from the grid. A conical horn launcher is proposed as an alternative for SW-TDR [62]. Use of a conical horn geometry has previously been bused for single wire in air communications [37]–[40]. It can be attached to the single wire without cutting it as well as isolating the launcher from the grounding grid. Fig. 13 shows the proposed conical horn launcher and its use. The dimensions of the horn launcher are $D_1 = 30$ mm, $D_2 = 200$ mm and $h = 100$ mm. The ground rod defect scenarios presented in Fig. 7 are investigated. Fig. 14 shows that the impulse response when using the horn launcher is comparable with the results obtained using ground plane-gap excitation. Apart from horn launchers, other surface wave launchers [32], [63]–[65] may also be suited to this application.

VII. SCALE MODEL SW-TDR EXPERIMENT

In this section a scale model experiment of the SW-TDR method is presented as a proof-of-concept of the technique. The experiment is conducted for ground rod scenarios A and C as shown in Fig. 7. Their length, distance to fault and break-point gap size are scaled by a 1/6 factor. The frequency and the bandwidth of the TDR input pulse is also scaled accordingly. Fig. 15 shows the scale model experimental setup. It consists of a 35 cm deep, 30 cm diameter container filled with sand. The launcher is constructed using a steel block with a 50Ω transition to interface the single wire ground rod with the TDR system. A screen ground mat is attached to the launcher which is completely embedded in the sand during the experiment. The TDR system produces a

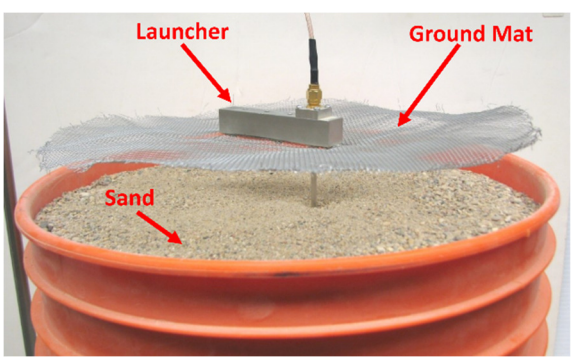
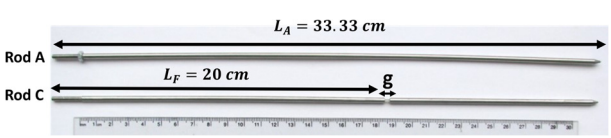
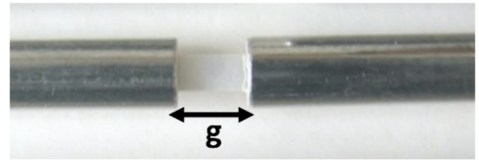


FIGURE 15: Single-Wire TDR experiment (1/6 scale). The wire (shown in Fig. 16) is connected to a 50Ω fixture which has a ground screen. The fixture and wire are entirely embedded in sand during measurements. The fixture is connected to a TDR instrument.

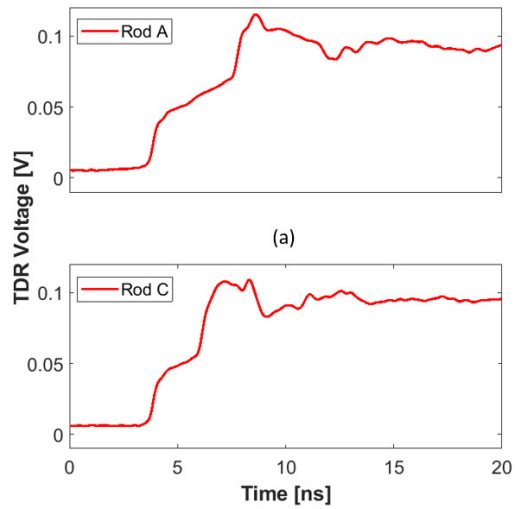


(a)



(b)

FIGURE 16. (a) Scaled versions of ground rod cases A and C (see Fig. 7). The diameter of each rod is 3.175 mm (b) Magnified view of the break-point of case C. The nylon gap spacer has a similar permittivity as sand.



(a)

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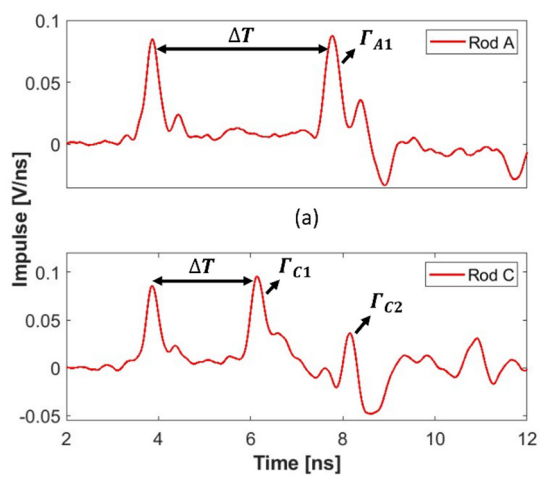
FIGURE 17: (a) TDR step responses for case A with no break-point and for (b) case C that has a break-point at 20 cm.



(a)

(b)

FIGURE 18: (a) TDR impulse responses for case A with no break-point and for (b) case C that has a break-point at 20 cm. The reflected signals show the same behavior as the simulated results for cases A and C in Fig. 9.

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TABLE II
CALCULATED AND MEASURED TIME DELAYS OF THE REFLECTED PULSE

	Break-point ΔT [ns]		End of Rod, ΔT [ns]	
	Theoretical	Measured	Theoretical	Measured
Rod A	n/a	n/a	3.91	3.92
Rod C	2.35	2.28	3.91	4.3

step input of amplitude 100 mV with a rise time of 200 ps (equivalent pulse bandwidth of 1.8 GHz). The electrical parameters of the sand are $\varepsilon_{rs} = 3.1$ and $\sigma_s = 0.0018$ S/m as measured using a TEM cavity resonator at 1.8 GHz. The scaled electrode is 33 cm long as in Fig. 16a with a break-point gap, g =3.3 mm, as shown in Fig. 16b made with nylon, which has a similar permittivity as sand. Two cases are investigated, one with and one without a break-point. Fig. 17 shows the measured TDR step responses and Fig. 18 shows the impulse responses for Rod cases A and C. The impulse responses are obtained from the derivative of the measured TDR step responses.

In the break-point free case the reflected pulse Γ_{A1} is due to the end of the rod. For the case with a break-point the reflected pulse Γ_{C1} is identified as due to the break-point located at 20 cm and Γ_{c2} is due to the end of the rod. These experiment results match with the simulated waveforms in Fig. 9. Note that, in the experiments, even though a very low amplitude TDR pulse with signal-noise-ratio (SNR) of 25 dB was used, a signal response capable of fault location detection was achieved. Creating TDR pulses with an amplitude 100 to 1000 times larger is easily achievable, greatly improving system SNR. Table II lists the theoretical and measured time delays of the reflected pulses due to a break-point and from the end of the rod. The small differences are due to the dispersion of the pulse and the fringing capacitance that is present at open-circuit discontinuities.

VIII. CONCLUSION

This paper presented a single wire TDR system and demonstrated it can be used for detecting faults in vertical grounding electrodes. Many fault scenarios were examined, showing the TDR approach can confidently detect break-points and determine their location. Additionally, intensity and shape of the TDR response waveform can be used to determine the severity and type of fault, including partially corroded, reduced radius, sections of the electrode.

A fast rise-time pulse is used such that its frequency bandwidth is high enough so that low dispersion propagation is achieved for a wide range of soil parameters. This enables accurate determination of discontinuities along the electrode, but it is accompanied by higher attenuation for higher conductivity soil. This limitation can be mitigated by using a high-amplitude TDR pulse.

A theory based on TM surface wave propagation was developed that matched the behaviour of the fields obtained from full wave simulation results when near the conductor. It is important to note that the radiated mode fields, generated at the source and at the open end of the conductor can also contribute to the total fields. All fields components are included in the full wave numerical simulations presented in this paper. To enhance the TM mode surface wave excitation use of a conical horn launcher was proposed which gave similar detection capabilities as the ground-gap excited launcher. The horn launcher provides a path to a method of exciting a ground electrode without the need to disconnect it from the grounding network.

A scale model experiment of the proposed SW-TDR system was demonstrated as a proof of concept. Measurements yielded outcomes comparable to the simulation results, validating the approach.

The proposed SW-TDR system is non-invasive, enabling detection of faults without removing the electrode or disconnecting it from the grounding network. By identifying a break-point or partial fault location and severity, the impedance degradation of an electrode can be predicted. This aids in determining the necessity of the electrode replacement. For instance, faults located near the end of the rod will have minimal impedance degradation, possibly negating the need for their replacement. Moreover, statistical analysis based on examination of a small number of electrodes can provide valuable insight for asset management and planning.

For grounding systems with very long electrodes or in environments with very high soil conductivity, the SW-TDR method will be less accurate due to increased attenuation of the reflected TDR pulse. In these situations, a high-amplitude TDR pulse can be used to provide improved signal-to-noise ratio. A lower frequency bandwidth pulse can also be used in these situations as attenuation will be lower, but at the expense of increased pulse dispersion. Additionally, the soil can be inhomogeneous, especially near the earth surface and on the surface of the electrode at corroded regions, and fault detection in such scenarios needs to be investigated further. It is expected that non-uniform corrosion layers on the surface of the electrode will not greatly affect TDR pulse propagation as it is primarily dominated by the soil properties. A more detailed analysis of the full TDR response signal, not just the first reflection as concentrated on in this paper, can be used to aid these investigations.

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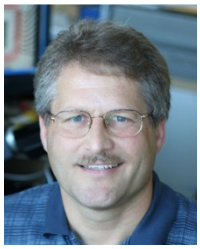

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